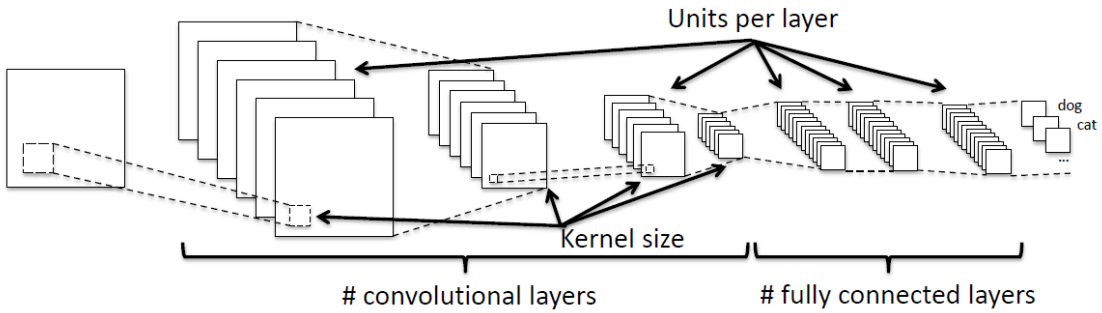
Hyper-parameter search

Mainly based on AutoML NeurIPS 2018 Tutorial
https://www.automl.org/events/

What are the hyper-parameters of a deep learning model?

- Any variable that you do not learn can (and ideally should) be treated as hyper-parameter
  - optimization choices
    - algorithm
      - first-order
      - 2nd-order
    - parameters of the algorithms
      - learning rate
      - momentum
      - batch size
  - architectural choices
    - batch normalization
    - dropout
  - parameters of loss function
    - weight decay
  - data augmentation techniques
  - …



Hyper-parameter Search

Motivation

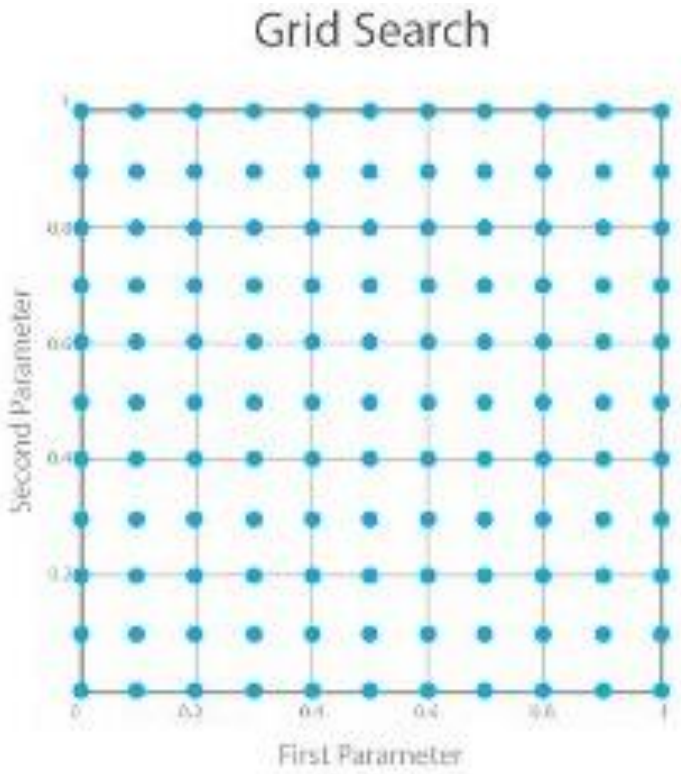
- performance of the model can *drastically* change from the worst to the best hyperparameters!
- Difference between one paper's model and the prior works might be explained by a more thorough HP search!
- Difference between one paper's model and the its baselines might be explained by a more thorough HP search! (baselines are commonly less optimized)

How can we do HP search?

Hyper-parameter Search

Grid Search

- Define a discretized interval for each HP
- Search by iterating through all possible combinations of HPs
- Most resource-hungry, but of course also most reliable in finding the optimum
- Simple implementation
- Easily parallelizable



Hyper-parameter Search

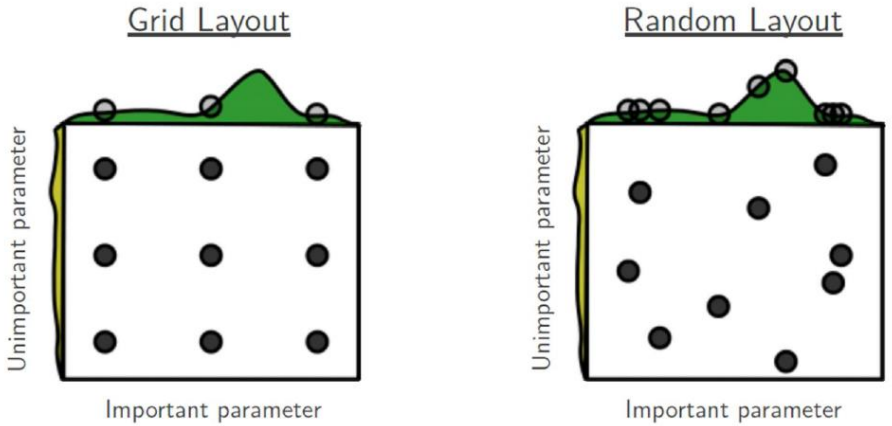
Random Search

- Randomly pick k points from the HP space (instead of equidistant grid points)
- Potentially more efficient than grid search but entails high variance

[Bergstra and Bengio. “Random Search for Hyper-Parameter Optimization”, JMLR 2012]

Random Search

- Good for the case of low effective dimensionality of HP space (e.g., batchsize vs learning rate)
- That is almost always the case in high-dimensional HP space (many hyper-parameters are correlated)



[Bergstra and Bengio. “Random Search for Hyper-Parameter Optimization”, JMLR 2012]

Random Search

For any distribution over a HP space with a finite maximum, the maximum of 60 random observations lies within the top 5% of possible configurations, with 95% probability.

$$1 - (1 - 0.05)^{60} \approx 0.954$$

Random Search

As you increase the number of (potentially correlating) hyperparameters, this top 5% might demonstrate less significant difference.

Random vs Grid Search

We've seen mixed result, what I would suggest is the following:

- If you have low number of HPs (relative to the available computational budget) use grid search

- When comparing multiple methods that require individual HP search, better to do a grid search to reduce the randomness of the comparison

Informed Search

- Where we base the selection of the current point (partially) based on the previously selected points and their performances
- Sequential Model Based Optimization (SMBO)
  - Take an initial set of random HP points
  - Get the performance of the model using the initial set
  - Train a HP model on these initial points and their performance
  - Sample a new (set of) point using the HP model
  - Get the performance of the models using the new set
  - Retrain/re-estimate the HP model and continue sampling

Hyper-parameter Search

Informed Search

- Where we base the selection of the current point (partially) based on the previously selected points and their performances
  - Gaussian Processes [Swerski et al. “Raiders of the Lost Architecture: Kernels for Bayesian Optimization in Conditional Parameter Spaces” NeurIPS workshop 2013]
  - Tree of Parzen Estimators (TPE) [Bergstra et al. “Algorithms for Hyper-Parameter Optimization”, NIPS 2011]
  - Genetic Algorithms [Loshchilov and Hutter, “CMA-ES FOR HYPERPARAMETER OPTIMIZATION OF DEEP NEURAL NETWORKS”, ICLR workshop 2016]
  - Multi-fidelity Optimization
  - Extrapolation of learning curves

HP Search – Hand tuning

Your know-how knowledge (is not to be undermined!)

- intervals of interest
- which parameters to include (e.g. not fine search along both batchsize and learning rate maybe?)
- Combination:
  - Coarse-level Random Search
  - Mid-level Bayesian Optimization
  - Fine-level grid-search
- …

After the HP search

use the whole training set to retrain

- we are assuming that the optimal HPs hold for the whole dataset
  - Usually size of the validation set is smaller than the training set
  - Ideally, use cross-validation for HP search